UNIVERSITY OF LAGOS

LECTURE NOTES

FLUID MECHANICS I

Properties of Fluids



Study Book

built from handwritten notes

Dark edition

Tap any line in Contents to jump to it

Contents

How to use this book **7**

1 Course Outline and Textbooks **10**

1.1 Course content 10

1.2 Textbooks 11

2 Fluid and Its Properties **13**

2.1 Definition of a fluid . 13

2.2 Fluid in motion and at rest 14

2.3 Characteristics of fluids 14

3 Density, Specific Volume, Specific Weight, Specific

Gravity **17**

3.1 Density or mass density 17

3.2 Specific volume . . . 18

3.3 Specific weight or weight density 19

3.4 Specific gravity or relative gravity 20

3.5 Quick checks (added) 22

4 Viscosity, Shear Stress and Newton's Law **25**

4.1 Viscosity 25

4.2 Shear stress 26

4.3 Coefficient of dynamic viscosity 29

4.4 Kinematic viscosity . . 30

4.5 Newton's law of viscosity 31

4.6 Worked checks (added) 33

5 Classification of Fluids 36

5.1 Classification of fluid . 36

5.2 A(i) Liquid 37

5.3 A(ii) Gas 37

5.4 A(iii) Vapour 38

5.5 B(i) Real or Viscous fluid 38

5.6 B(ii) Ideal or Inviscid fluid 40

6 Newtonian and Non-Newtonian Fluids 45

6.1 Newtonian fluids . . . 45

6.2 Non-Newtonian fluids 47

6.3 Plastic fluids and thixotropic fluid . . . 48

6.4 Pseudoplastic and dilatant fluids 50
6.5 Non-Newtonian classification 52

7 Formula and Units Sheet 60

8 Comparison Tables 64

8.1 Newtonian vs Non-Newtonian 64
8.2 Real vs Ideal fluid . . 65
8.3 Liquid, gas, vapour . . 66
8.4 Types of non-Newtonian fluid . . . 67

9 Transcription Checks 69

9.1 Handwriting readings . 69
9.2 Spelling tidied 71

9.3 Items placed out of order 72

9.4 Notes that disagree with themselves . . . 73

9.5 What was added . . . 74

10 Symbols Index 75

How to use this book

Every page has a link at the bottom: ↑ Contents takes you back to the contents page.

Colour key

> **Grey bar**: your notes, word for word.

Definition

From your notes.

Formula

From your notes.

Note on your notes

Your notes disagree with themselves or are unclear. Your version is kept.

Added clarification

Extra science, not in your notes.

Key takeaways (added)

Section summary.

Common mistakes (added)

Errors to avoid.

Exam tip (added)

How to answer.

Memory tricks (ours)

Our own memory tricks. Not from your notes.

Page tags. [p. 4] means "from page 4 of your notebook".

Reading order. Where a sentence ran across a page, it is joined here. The notes on *Ideal or Inviscid fluid*

(written on page 13) sit next to *Real or Viscous fluid* in Section 5 so the two can be compared.

Spelling. Small spelling slips were tidied. Every change is listed in Section 9.

1 Course Outline and Textbooks

1.1 Course content

[p. 1]

> **FLUID MECHANICS I**
> Course content:
> 1) Real and Ideal fluids
> 2) Differentiation following the motion of fluid particles
> 3) Equation of continuity and applications
> 4) Equation of motion for incompressible Inviscid fluids
> 5) Velocity potentials and Stoke's stream functions
> 6) Bernoulli's equation with appli-

cations

7) Kinetic energy
8) Sources, Sinks and doublets in two or three dimensions
9) Streamlines and Images
10) Use of Conformal transformation

1.2 Textbooks

[p. 1–2]

Textbooks

1) A textbook of fluid mechanics by R.K. Rajput & S. Chand, S. Chand & Company Ltd, Ramnagar, New Delhi
2) Introductory lectures on fluid dynamics by Roger. K. Smith
3) Any textbook on fluid mechanics.

Key takeaways (added)

- Ten topics, starting with *Real and Ideal fluids*.
- This book covers fluid properties, viscosity, and the split between real and ideal fluids.

2 Fluid and Its Properties

2.1 Definition of a fluid

[p. 3]

> **FLUID AND ITS PROPERTIES**
>
> Definition: A fluid is a substance which is capable of flowing.
>
> MTH DEF: A fluid can also be defined as a substance which deforms continuously when subjected to external shearing force.

Fluid

A substance which is **capable of flowing**.
Or: a substance which **deforms continuously** when subjected to ex-

ternal shearing force.

2.2 Fluid in motion and at rest

[p. 3]

> When fluid is in motion, there is a shear stress or force acting on it at that time, but if the fluid is at rest, there is no shear stress or force acting on it.

2.3 Characteristics of fluids

[p. 3]

> Characteristics of fluids:
> A fluid has the following characteristics:
>
> 1 It has no definite shape, but con-

forms to the shape of the containing vessel.

2 A small amount of shear stress exerted on the fluid causes a continuous deformation on the fluid as long as the force is applied.

Key takeaways (added)

- Fluid = capable of flowing = deforms continuously under shearing force.
- In motion: shear stress acts. At rest: no shear stress.
- No definite shape. It takes the shape of its container.
- Even a small shear stress keeps it deforming while the force lasts.

Common mistakes (added)

- Mixing up *shearing* force with *normal* force (pressure).
- Thinking a fluid needs a large force before it starts to deform.

Exam tip (added)

Write both definitions: "capable of flowing" and "deforms continuously under external shearing force".

Memory tricks (ours)

Fluid = soup in a pot.

It takes the **shape of the pot**: no definite shape.

Stir it (shear) and it keeps flowing: continuous deformation.

Stop stirring (rest) and there is no shear.

3 Density, Specific Volume, Specific Weight, Specific Gravity

3.1 Density or mass density

[p. 4]

Properties of fluids

1) Density or mass density

Fluid density is defined as the ratio of Mass of the fluid to its volume. Mathematically,

$$\text{Density} = \rho = \frac{\text{Mass of the fluid}}{\text{volume of the fluid}}$$

Its SI unit is $\mathrm{kg\,m^{-3}}$.

The value of density of water is $1000\ \mathrm{kg\,m^{-3}}$.

Density

$$\rho = \frac{\text{mass}}{\text{volume}}$$

SI unit: $\mathrm{kg\,m^{-3}}$

Water: $1000\ \mathrm{kg\,m^{-3}}$

3.2 Specific volume

[p. 4]

2) Specific Volume
The specific volume of a fluid is the ratio of the volume of the fluid to the mass of the fluid.
Mathematically:

$$\text{Specific volume} = \frac{\text{volume of the fluid}}{\text{mass of the fluid}}$$

It is the reciprocal of mass density and it is commonly applied for gases. Its SI unit is $\mathrm{m^3\,kg^{-1}}$.

Specific volume

$$\text{specific volume} = \frac{\text{volume}}{\text{mass}} = \frac{1}{\rho}$$

SI unit: $\mathrm{m}^3\,\mathrm{kg}^{-1}$

Added clarification

Your notes say specific volume is *commonly applied for gases*. The idea itself (volume per unit mass) works for any substance.

3.3 Specific weight or weight density

[p. 4–5]

> 3 Specific weight or weight density
> The specific weight or weight density of a fluid is the ratio of the weight of a fluid to its volume.

Mathematically,
Specific weight (w)

$$= \frac{\text{weight of the fluid}}{\text{volume of the fluid}}$$
$$= \frac{\text{Mass of the fluid} \times g}{\text{volume of the fluid}}$$

$$w = \rho g$$

The value of specific weight for water is $9.8 \times 1000\ \mathrm{N\,m^{-3}}$.

Specific weight

$$w = \frac{\text{weight}}{\text{volume}} = \rho g$$

SI unit: $\mathrm{N\,m^{-3}}$

Water: $9.8 \times 1000\ \mathrm{N\,m^{-3}}$

3.4 Specific gravity or relative gravity

[p. 5]

4 Specific Gravity or relative gravity

The specific gravity or relative gravity of a fluid is defined as the ratio of density of the fluid to ratio of density of standard fluid at a specified temperature. For liquids, the standard fluid is taken as water. For gases, the standard fluid is taken as air.

Specific gravity

$$\text{SG} = \frac{\rho_{\text{fluid}}}{\rho_{\text{standard fluid}}}$$

Liquids: standard fluid = water
Gases: standard fluid = air

Added clarification

A ratio of two densities has no unit (added). Your notes do not state this.

3.5 Quick checks (added)

Worked checks (added)

1. Gas with $\rho = 1.25\ \mathrm{kg\,m^{-3}}$.
specific volume $= \dfrac{1}{1.25} = 0.8\ \mathrm{m^3\,kg^{-1}}$

2. Fluid with $\rho = 850\ \mathrm{kg\,m^{-3}}$, $g = 9.8$.
$w = \rho g = 850 \times 9.8 = 8330\ \mathrm{N\,m^{-3}}$

3. Fluid with SG $= 0.8$ and water at $1000\ \mathrm{kg\,m^{-3}}$.
$\rho = 0.8 \times 1000 = 800\ \mathrm{kg\,m^{-3}}$

Key takeaways (added)

- ρ = mass ÷ volume. Water: 1000.
- Specific volume $= 1/\rho$.
- $w = \rho g$. Water: 9.8×1000.
- SG compares a density to water

(liquids) or air (gases).

Common mistakes (added)

- Weight ÷ volume gives specific weight, not density. Divide by g to reach density.
- Treating specific gravity as a density with units.
- Multiplying by g instead of taking the reciprocal for specific volume.

Memory tricks (ours)

Density = a danfo. Mass is the **people**, volume is the **space**. Packed danfo = high ρ.

Specific volume = elbow room. Space per person, $1/\rho$. Packed danfo = no elbow room.

Weight density: $w = \rho g$**.** "Gravity weighs the crowd."

SG = class representative. Liquids pick **Water**, gases pick **Air**. It is a position in class, so it has **no unit**.

Water 1000: 1 m^3 is 1000 litres, and a litre of water is about 1 kg. Times g gives 9.8×1000.

4 Viscosity, Shear Stress and Newton's Law

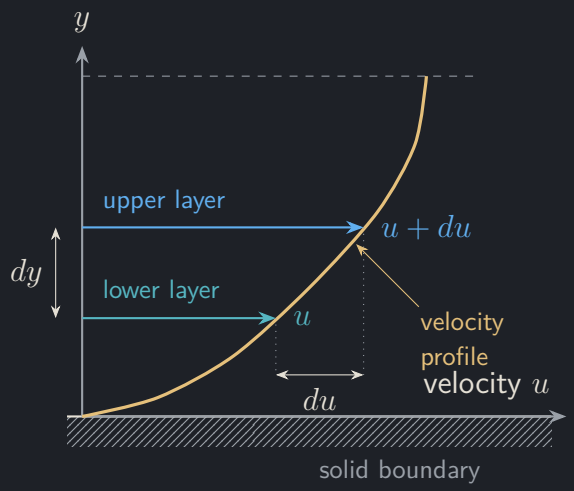
4.1 Viscosity

[p. 5–6]

> **Viscosity**
> Viscosity is defined as the property of fluid which offers resistance to the movement of one layer of the fluid over another adjacent layer. When two layers of a fluid at a distance dy move over each other with velocities u and $u + du$ as shown in the diagram below

Viscosity

The property of fluid which offers **resistance** to the movement of one layer of the fluid over another adjacent layer.

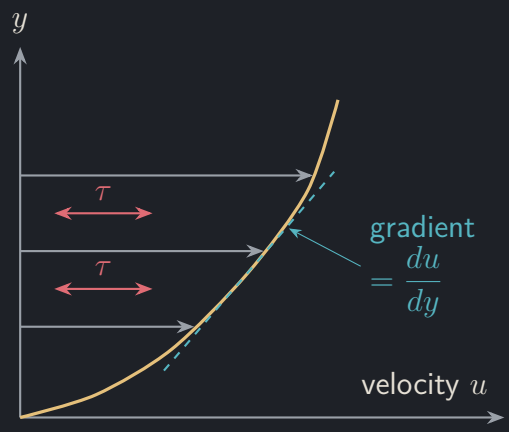


Velocity variation near solid boundary [p. 6]

4.2 Shear stress

[p. 7]

> The viscosity together with the relative velocity causes a force to act between the two layers, and this force is called the shear stress or the shear force, denoted by τ or τ_{xy} (for two-dimensional space).
> The top layer causes a shear stress on the adjacent lower layer while the lower layer causes a shear stress on the adjacent top layer as shown in the diagram below



Shear stress between layers and the velocity gradient [p. 7]

> $du/dy =$ rate of shear strain
> $\tau =$ Shear stress
> Hence the shear stress τ is directly proportional to the rate of shear strain (du/dy) or the velocity gradient in the direction perpendicular to

the layers. Therefore

$$\tau \propto \frac{du}{dy}$$

Shear stress law

$$\tau = \mu \frac{du}{dy}$$

$\frac{du}{dy}$ = rate of shear strain (velocity gradient)
τ = shear stress

4.3 Coefficient of dynamic viscosity

[p. 8]

$$\tau = \mu \frac{du}{dy}$$

Where μ is the constant of proportionality called the viscosity or coef-

ficient of dynamic viscosity.

$$\mu = \frac{\tau}{du/dy} = \text{Viscosity}$$

The SI unit of viscosity (μ) is $\mathrm{N\,s\,m^{-2}}$ (Newton seconds per meter square)

Dynamic viscosity

$$\mu = \frac{\tau}{du/dy}$$

SI unit: $\mathrm{N\,s\,m^{-2}}$

4.4 Kinematic viscosity

[p. 8]

Definition
Kinematic Viscosity ν
Kinematic viscosity ν is defined as the ratio between dynamic viscosity

(μ) and the density of the fluid i.e

$$\nu = \frac{\mu}{\rho}$$

its SI unit is $\mathrm{m^2 s^{-1}}$

Kinematic viscosity

$$\nu = \frac{\mu}{\rho}$$

SI unit: $\mathrm{m^2\,s^{-1}}$

4.5 Newton's law of viscosity

[p. 8–9]

NEWTON'S LAW OF VISCOSITY

It states that the shear stress (τ) on a fluid element layer is directly proportional to the velocity gradient

or the rate of shear strain.

$$\tau \propto \frac{du}{dy}$$

$\tau \rightarrow$ shear stress
$du/dy \rightarrow$ velocity gradient or rate of shear strain

where

$$\tau = \mu \frac{du}{dy}$$

where μ is the constant of proportionality called the coefficient of dynamic viscosity or simply viscosity

Newton's law of viscosity

Shear stress on a fluid element layer is directly proportional to the velocity gradient (rate of shear strain).

$$\tau = \mu \frac{du}{dy}$$

Added clarification

Units (added). Shear strain itself has no unit. The *rate* of shear strain du/dy is in s^{-1}. That is why $\tau = \mu\, du/dy$ gives $\mathrm{N\,s\,m^{-2}} \times \mathrm{s^{-1}} = \mathrm{N\,m^{-2}}$.

4.6 Worked checks (added)

Worked checks (added)

1. Shear stress. $\mu = 0.048\ \mathrm{N\,s\,m^{-2}}$. Velocity $1.125\ \mathrm{m/s}$ at $75\ \mathrm{mm}$ from the wall, linear profile.

$$\frac{du}{dy} = \frac{1.125}{0.075} = 15\ \mathrm{s^{-1}}$$

$$\tau = 0.048 \times 15 = 0.72\ \mathrm{N\,m^{-2}}$$

2. Kinematic viscosity. $\mu = 0.003$, $\rho = 800$.

$$\nu = \frac{0.003}{800} = 3.75 \times 10^{-6}\ \mathrm{m^2\,s^{-1}}$$

Key takeaways (added)

- Viscosity $=$ resistance to one layer moving over another.
- Layers at distance dy move at u and $u + du$.
- $\tau \propto du/dy$, so $\tau = \mu\, du/dy$.
- μ unit: $\mathrm{N\,s\,m^{-2}}$. $\nu = \mu/\rho$, unit $\mathrm{m^2 s^{-1}}$.

Common mistakes (added)

- Mixing the units of μ ($\mathrm{N\,s\,m^{-2}}$) and ν ($\mathrm{m^2 s^{-1}}$).
- Forgetting to change mm to m in dy.
- Thinking τ is proportional to u itself. It is proportional to the *gradient* du/dy.
- Dividing ρ by μ instead of μ by ρ.

Exam tip (added)

Layout for calculations: Definition $\rightarrow$ Formula $\rightarrow$ Substitution $\rightarrow$ Units $\rightarrow$ Final answer.

Memory tricks (ours)

Stress = Sticky $\times$ Steep.

$\tau = \mu\, du/dy$: μ is the **stickiness** (garri paste, honey), du/dy is how **steeply** the layers slide.

Unit of μ: τ is $\mathrm{N\,m^{-2}}$. You divide by a rate that is "per second", so μ gains a **second**: $\mathrm{N\,s\,m^{-2}}$.

$\nu = \mu/\rho$: "**Moo** over **Row** = **New**" (μ over ρ gives ν).

Dynamic has Newtons (force). **Kinematic has no force**, only length and time: $\mathrm{m^2\,s^{-1}}$.

5 Classification of Fluids

5.1 Classification of fluid

[p. 9]

Classification of fluid:
A fluid may be classified as follows

A) (i) liquid (ii) Gas and (iii) vapour

B) (i) Real or Viscous fluid (ii) Ideal or Inviscid fluid

Two ways to classify

A) By state: liquid, gas, vapour.
B) By viscosity: real (viscous) fluid, ideal (inviscid) fluid.

5.2 A(i) Liquid

[p. 9]

> A(i) Liquid is a fluid with definite volume which varies slightly with temperature and pressure. It is assumed to be incompressible but undergoes viscosity.
> Liquid vaporises at narrow pressure above 0 depending on the temperature

Unclear phrase in your notes

"narrow pressure above 0" is kept exactly as written. Its meaning is not clear from the handwriting. Please check it against your lecturer's wording or textbook.

5.3 A(ii) Gas

[p. 10]

> A(ii) Gas does not have a definite volume, but it is compressible. Gas completely fills the vessel which contains it.

5.4 A(iii) Vapour

[p. 10]

> A(iii) Vapour is a gas whose temperature and pressure are very near the steam or liquid state.

5.5 B(i) Real or Viscous fluid

[p. 10]

> B(i) Real or Viscous fluids are actually available in nature. They have

viscosity, surface tension, density and are compressible. Since they have viscosity i.e $\mu > 0$, they are called real fluids and their motion is known as viscous flow e.g gases, air, water, kerosine etc.

Real or viscous fluid is further classified as Newtonian and Non-Newtonian fluids.

Two statements to hold together

Page 9: a liquid "is assumed to be incompressible".

Page 10: real fluids "are compressible".

Both are kept as written. In the notes, treat a liquid as incompressible in calculations (p. 9), and a real fluid in general as compressible (p. 10).

5.6 B(ii) Ideal or Inviscid fluid

[p. 13]

B(ii) Ideal or Inviscid fluid are fluids that do not exist in nature. They have no viscosity $\mu = 0$, no surface tension, they are irrotational and are incompressible.
In an ideal flow, there is no existence of shear force because of vanishing viscosity i.e

$$\tau = \mu \frac{du}{dy} = 0 \quad \text{since } \mu = 0$$

The shear stress or force τ is always zero regardless of the fluid motion. Since ideal fluid has no viscosity, they are called ideal fluids and their resulting motion is called an ideal or inviscid flow

Ideal flow

$$\tau = \mu \frac{du}{dy} = 0 \quad \text{since } \mu = 0$$

Added clarification

Three separate ideas (added). *Inviscid* means $\mu = 0$. *Incompressible* means constant density. *Irrotational* means no spinning of fluid particles. Your notes give an ideal fluid all three, but they are different properties.

Key takeaways (added)

- Liquid: definite volume, assumed incompressible, has viscosity.
- Gas: no definite volume, compressible, fills its vessel.

- Vapour: a gas very near the steam or liquid state.
- Real: $\mu > 0$, surface tension, density, compressible. Viscous flow.
- Ideal: $\mu = 0$, no surface tension, irrotational, incompressible. Does not exist. $\tau = 0$ always.

Common mistakes (added)

- Calling vapour a completely different state. It is a gas close to liquid.
- Saying ideal fluids exist in a laboratory. They do not exist in nature.
- Thinking $\tau = 0$ in ideal flow is because the fluid is not moving. It is because $\mu = 0$.

Exam tip (added)

Write the real-fluid list in the notes' order: viscosity, surface tension, density, compressible.

Memory tricks (ours)

LGV = "**L**agos **G**ives **V**ibes": liquid, gas, vapour.

Liquid = bottled water, keeps its volume.

Gas = perfume, fills the whole room.

Vapour = steam over the boiling pot, a gas about to turn liquid.

Real = V-S-D-C = "**V**ery **S**erious **D**anfo **C**onductor". A real Lagos conductor exists in nature: **V**iscosity, **S**urface tension, **D**ensity, **C**ompressible.

Ideal = a 0–0 draw. Zero viscosity, **Zero** surface tension, then **I**rrotational and **I**ncompressible ("I, I").

An ideal fluid is like **24-hour light in Nigeria**: lovely idea, does not exist in nature.

6 Newtonian and Non-Newtonian Fluids

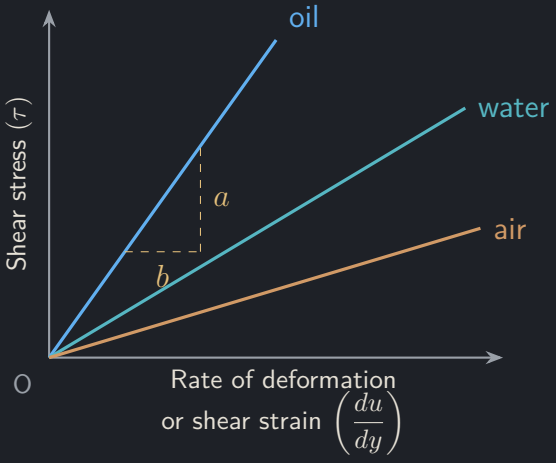
6.1 Newtonian fluids

[p. 10–11]

> What are Newtonian fluids?
> Newtonian fluids are fluids that follows newton's law of viscosity. In Newtonian fluid, the fluid viscosity does not change with the rate of deformation. The shear stress is linearly proportional to the velocity gradient or rate of shear strain. The plot of the shear stress against the velocity gradient or rate of deformation is a straight line through the origin. The slope of the line

determines the viscosity. In one dimensional shear flow of Newtonian fluids, the shear stress can be expressed by the linear relationship

$$\tau = \mu \frac{du}{dy}$$



Newtonian fluids: shear stress against rate of deformation [p. 11]

Newtonian fluid

$$\tau = \mu \frac{du}{dy}$$

Straight line through the origin. The slope determines the viscosity.

6.2 Non-Newtonian fluids

[p. 11–12]

> Non-Newtonian fluids
> These fluids do not obey newton's law of viscosity or Newton Viscosity equation. This is because they do not follow the linear relationship between the shear stress and the rate of deformation. These fluids are relatively not common. e.g solutions or Suspensions (slurries), mud flows, polymer solutions, blood etc.
> These fluids are generally complex

mixtures and are studied under rheology which is a science of deformation and flow.

Rheology

The science of deformation and flow. Non-Newtonian fluids are studied under it.

6.3 Plastic fluids and thixotropic fluid

[p. 12]

Other non Newtonian fluids are plastic fluids e.g Ideal plastics or Bingham plastic. These have definite yield stress and a constant linear relation between the shear stress and

the rate of angular deformation i.e

$$\tau = \text{Constant} + \mu \frac{du}{dy}$$

Examples are Sewage, sludge, drilling mud, water Suspension of clay & flyash.

A non Newtonian thixotropic fluid has a nonlinear relationship between the shear stress and the rate of angular deformation beyond an initial yield stress i.e

$$\tau = \text{Constant} + \mu \left(\frac{du}{dy}\right)^n$$

Example is printer ink.
For Non Newtonian fluid,

$$\tau = \left(\frac{du}{dy}\right)^n,$$

Note: thixotropic equation

Page 12 gives the thixotropic fluid as $\tau = \text{Constant} + \mu(du/dy)^n$. Page 14 gives it as $\tau = \mu(du/dy)^n + f(t)$ with $f(t)$ decreasing. Both are kept as written. Use the page 14 form for the time-dependent table.

Note: missing μ

The last line on page 12, $\tau = (du/dy)^n$, has no μ in front. It is kept as written. Pages 13–14 write $\tau = \mu(du/dy)^n$.

6.4 Pseudoplastic and dilatant fluids

[p. 13]

when $n < 1$, the fluid is called pseudo plastic fluid e.g paper pulp, rubber suspension paints, blood, milk, liquid cement and clay.
When $n > 1$, the fluid is called Dilatant fluid e.g butter, printing inks, sugar solution, aqueous suspension.

Power form

$$\tau = \mu \left(\frac{du}{dy}\right)^n$$

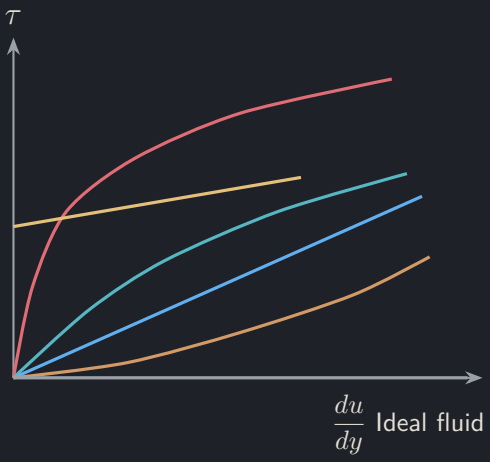
$n < 1$: pseudoplastic
$n > 1$: dilatant

Note: printer ink

Page 12 and page 14 list *printer ink* under thixotropic. Page 13 lists *printing inks* under dilatant. Both are kept as written.

6.5 Non-Newtonian classification

[p. 14]



Rheology curves of fluid types [p. 14]

Non-Newtonian fluids $\tau \neq \mu \frac{du}{dy}$

Purely viscous fluid: Time-independent

1) pseudo plastic fluid

$$\tau = \mu \left(\frac{du}{dy}\right)^n, \ n < 1$$

Exp: Blood, milk

2) Dilatent fluid

$$\tau = \mu \left(\frac{du}{dy}\right)^n, \ n > 1$$

Example: Butter

3) Bingham or Ideal plastic fluid

[p. 15]

$$\tau = \tau_0 + \mu \left(\frac{du}{dy}\right)^n$$

Example: water suspension of clay & flyash.

Purely viscous fluid: Time dependent

Thixotropic fluid

$\tau = \mu \left(\frac{du}{dy}\right)^n + f(t)$

$f(t)$ is decreasing

Exp: printer ink, Crude oil.

Rheopectic fluid

$\tau = \mu \left(\frac{du}{dy}\right)^n + f(t)$

$f(t)$ is increasing

Example: Rare liquid Solid Suspension

Visco-elastic fluid

Visco-elastic fluids

$\tau = \mu \frac{du}{dy} + \alpha E$

Example: Liquid – solid Combinations in pipe flow.

Note: Bingham exponent

Page 12 writes the Bingham plastic as linear: $\tau = \text{Constant} + \mu\, du/dy$. Page 15 writes it with an exponent: $\tau = \tau_0 + \mu(du/dy)^n$.
Both are kept as written. Use page 12 for the straight-line relation, and page 15 for the table form.

Note: αE

α and E are not defined in your notes. Kept as written.

Added clarification

Time-independent vs time-dependent (added). Pseudoplastic and dilatant behaviour depends on the *rate* of shear. Thixotropic and rheopectic behaviour also depends on *time* under shear. That is why those two carry $f(t)$.

Key takeaways (added)

- Newtonian: $\tau = \mu\, du/dy$. Straight line through the origin.
- Non-Newtonian: $\tau \neq \mu\, du/dy$. Studied under rheology.
- Pseudoplastic $n < 1$. Dilatant $n > 1$.
- Bingham plastic: definite yield stress, then a linear relation.
- Thixotropic: $f(t)$ decreasing. Rheopectic: $f(t)$ increasing.
- Visco-elastic: $\tau = \mu\, du/dy + \alpha E$.

Common mistakes (added)

- Mixing $n < 1$ (pseudoplastic) with $n > 1$ (dilatant).
- Mixing thixotropic ($f(t)$ decreasing) with rheopectic ($f(t)$ in-

creasing).

- Drawing Bingham plastic from the origin. It starts at a yield stress on the τ axis.
- Putting visco-elastic fluids in the purely viscous group.

Exam tip (added)

For the table, give: name, equation, condition (n or $f(t)$), example. Use the examples in your notes.

Memory tricks (ours)

Newtonian = Neat. A **neat straight line** through the origin. The **slope is the stickiness** (μ).

Rheology = the science of flow ("rheo" means flow).

Less than 1, less thick. Pseudo-

plastic ($n < 1$) thins when pushed: **P**aint, **M**ilk, **B**lood.

Greater than 1, greater grip. Dilatant ($n > 1$) digs in and thickens when pushed. Think cornflour paste. Butter is the notes' example.

Bingham = bouncer. Pay the **entry fee** (τ_0, the yield stress) first, then it flows. Think toothpaste.

Thixotropic = **T**hins with **T**ime: $f(t)$ **decreasing**.

Rheopectic = **R**ises with time: $f(t)$ **increasing**.

Visco-elastic = viscous **plus** elastic: $\mu\, du/dy + \alpha E$, like pounded yam that flows and bounces back.

Time-independent are **P-D-B** (pseudo, dilatant, Bingham). Only

thixotropic and rheopectic carry the clock, $f(t)$.

7 Formula and Units Sheet

Each card is from your notes. Tap the section number under a card to jump back to it.

Density

$$\rho = \frac{\text{mass}}{\text{volume}}$$

SI unit: $\mathrm{kg\,m^{-3}}$

§3.1 Water: $1000\ \mathrm{kg\,m^{-3}}$

Specific volume

$$\frac{\text{volume}}{\text{mass}} = \frac{1}{\rho}$$

SI unit: $\mathrm{m^3\,kg^{-1}}$

§3.2

Specific weight

$$w = \frac{\text{weight}}{\text{volume}} = \rho g$$

SI unit: $\mathrm{N\,m^{-3}}$

§3.3 Water: 9.8×1000

Specific gravity

$$\text{SG} = \frac{\rho_{\text{fluid}}}{\rho_{\text{standard}}}$$

SI unit: none (a ratio, added)

§3.4 Liquids: water. Gases: air.

Newton's law of viscosity

$$\tau = \mu \frac{du}{dy}$$

SI unit: τ in $\mathrm{N\,m^{-2}}$ (added)

§4.5

Dynamic viscosity

$$\mu = \frac{\tau}{du/dy}$$

SI unit: $\mathrm{N\,s\,m^{-2}}$

§4.3

Kinematic viscosity

$$\nu = \frac{\mu}{\rho}$$

SI unit: $\mathrm{m^2\,s^{-1}}$

§4.4

Ideal flow

$$\tau = \mu \frac{du}{dy} = 0 \text{ since } \mu = 0$$

SI unit: –

§5.6

Pseudoplastic ($n < 1$) / Dilatant ($n > 1$)

$$\tau = \mu \left(\frac{du}{dy} \right)^n$$

SI unit: –

§6.4

Bingham plastic

$\tau = \text{Constant} + \mu \dfrac{du}{dy}$ (p. 12)

$\tau = \tau_0 + \mu \left(\dfrac{du}{dy}\right)^n$ (p. 15)

SI unit: –

§6.3, §6.5

Thixotropic / Rheopectic

$\tau = \mu \left(\dfrac{du}{dy}\right)^n + f(t)$

Thixotropic: $f(t)$ decreasing

Rheopectic: $f(t)$ increasing

SI unit: –

§6.5

Visco-elastic

$\tau = \mu \dfrac{du}{dy} + \alpha E$

SI unit: –

§6.5

8 Comparison Tables

All entries come from your notes. A dash means the notes do not say.

8.1 Newtonian vs Non-Newtonian

	Newtonian	Non-Newtonian
Newton's law	Obeyed	Not obeyed
τ vs du/dy	Linear. Straight line through origin	Not linear
Equation	$\tau = \mu \frac{du}{dy}$	$\tau \neq \mu \frac{du}{dy}$
Examples	oil, water, air (p. 11 plot)	slurries, mud flows, polymer solutions, blood

8.2 Real vs Ideal fluid

	Real or Viscous	Ideal or Inviscid
In nature	Actually available	Do not exist
Viscosity	$\mu > 0$	$\mu = 0$
Surface tension	Yes	None
Compressible	Yes	Incompressible
Irrotational	–	Yes
Shear stress	$\tau = \mu \frac{du}{dy}$	Always zero
Motion called	Viscous flow	Ideal or inviscid flow
Examples	gases, air, water, kerosine etc.	–

8.3 Liquid, gas, vapour

	Volume	Other
Liquid	Definite. Varies slightly with temperature and pressure	Assumed incompressible. Has viscosity
Gas	Not definite	Compressible. Fills its vessel
Vapour	–	A gas very near the steam or liquid state

8.4 Types of non-Newtonian fluid

Type	Equation and condition	Example
Pseudo plastic	$\tau = \mu(\frac{du}{dy})^n$, $n < 1$	Blood, milk
Dilatant	$\tau = \mu(\frac{du}{dy})^n$, $n > 1$	Butter
Bingham or Ideal plastic	$\tau = \tau_0 + \mu(\frac{du}{dy})^n$	Water suspension of clay & flyash
Thixotropic	$\tau = \mu(\frac{du}{dy})^n + f(t)$, $f(t)$ decreasing	Printer ink, Crude oil
Rheopectic	$\tau = \mu(\frac{du}{dy})^n + f(t)$, $f(t)$ increasing	Rare liquid Solid Suspension
Visco-elastic	$\tau = \mu\frac{du}{dy} + \alpha E$	Liquid – solid combinations in pipe flow

Added clarification

Two different ideas (added). $n < 1$ and $n > 1$ describe how stress changes with the *rate* of shear. $f(t)$ describes change with *time*.

9 Transcription Checks

Every place where the handwriting was read by judgement, or where a small change was made, is listed. Please check these against your notebook.

9.1 Handwriting readings

1. [p. 1] Item 5 has a struck-out word before "stoke's stream functions". It was left out: "Velocity potentials and Stoke's stream functions".
2. [p. 1] The sign "$" is used as an ampersand in "R.K. Rajput $ S. Chand" and "S. Chand $ company ltd". Written here as "&".
3. [p. 3] "MTH DEF:" is kept as written. What "MTH" stands for is not stated.

4. [p. 4–5] Item numbers 3 and 4 at the left margin are kept as item numbers.
5. [p. 4–5] The unit of the specific weight of water, written like "Nm^{-s}", is read as $\mathrm{N\,m^{-3}}$.
6. [p. 5] "Specific Granty or relative granty" is read as "Specific Gravity or relative gravity".
7. [p. 5] "ratio of density of the fluid to ratio of density of standard fluid" is kept exactly as written, including the second "ratio".
8. [p. 7] "τ or τ_{xy}" is read as written with xy as a subscript.
9. [p. 9] "narrow pressure above 0" is kept as written. The meaning is unclear (see the note in §5.2).
10. [p. 10] "kerosine" is kept as written.

11. [p. 11] On the plot, “a” and “b” on the slope triangle are read as the vertical and horizontal sides.
12. [p. 14] The plot has no label on the vertical axis. τ was assumed, because it is the quantity plotted against du/dy in the notes.
13. [p. 14] The visco-elastic equation reads like “$\mu\, dy/dy + \alpha E$” and is read as $\mu\, du/dy + \alpha E$. α and E are not defined in the notes.
14. [p. 14–15] Page 15 item 3 (Bingham) continues the time-independent column of the page 14 table, from the table lines drawn on page 15.

9.2 Spelling tidied

1. [p. 12] “thyxotropic” → “thixotropic”.

2. [p. 13] “Dilatent” → “Dilatant”. The page 14 table keeps “Dilatent fluid” as written.
3. [p. 9,13] “Invicid” → “Inviscid”.
4. [p. 1] “Differentation” → “Differentiation”.

9.3 Items placed out of order

- Ideal or Inviscid fluid (page 13) sits in §5.6, next to Real fluid.
- Newton’s law (starts at the bottom of page 8, ends on page 9) is joined in §4.5.

9.4 Notes that disagree with themselves

All kept as written, each with a "Note on your notes" box:

- Thixotropic equation, p. 12 vs p. 14 (§6.3).
- Missing μ on the last line of p. 12 (§6.3).
- Printer ink, thixotropic vs dilatant (§6.4).
- Bingham exponent, p. 12 vs p. 15 (§6.5).
- Liquid "incompressible" vs real fluid "compressible" (§5.5).
- Undefined αE (§6.5).

9.5 What was added

Everything in the teal, red, gold, purple and grey boxes is added and is not from your notes. The blue, green and orange boxes follow your notes. The abbreviation SG and the unit statements marked "added" are also additions.

10 Symbols Index

ρ Density or mass density. §3.1

w Specific weight or weight density. §3.3

g Used in $w = \rho g$. §3.3

SG Specific gravity (abbreviation added). §3.4

μ Viscosity, coefficient of dynamic viscosity. §4.3

ν Kinematic viscosity. §4.4

τ Shear stress or shear force. §4.2

τ_{xy} Shear stress in two-dimensional space. §4.2

u Velocity. §4.1

y Distance perpendicular to the layers. §4.1

dy Distance between two layers. §4.1

du Velocity difference of two layers. §4.1

$\frac{du}{dy}$ Rate of shear strain, velocity gradient. §4.2

n Power in $\tau = \mu(du/dy)^n$. §6.4

τ_0 Constant term in the Bingham plastic. §6.5

$f(t)$ Time function, thixotropic and rheopectic. §6.5

α, E Visco-elastic terms, not defined in the notes. §6.5

End of book. Questions and quizzes will come in a later edition.